Graphic Era University

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Railway Reservation and Waiting List Management System

Student / Team Information

Team Name / Number	Team DSCPP-III-2026-T300
Team member 1 (Team Lead) (Name, ID, email, photo) 	Anmol Pandey – 2510010778 anmolpandey2122@gmail.com
Team member 2 (Name, ID, email, photo) 	Prakul Maithani – 2510011549 maithaniprakul373@gmail.com

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

(Brief restatement of your project's main goal. Max 300 words).

The Railway Reservation and Waiting List Management System is designed to simulate the basic operations of a railway reservation counter using C++.Users will be able to view train and seat information, book tickets, cancel confirmed reservations, search passenger details, and view confirmed passengers, the waiting list, and cancellation history.A linked list will manage confirmed passenger records,a queue will maintain waiting passengers in First-In-First-Out (FIFO) order,and a stack will store cancellation history in Last-In-First-Out (LIFO) order.The system will use classes such as `Passenger`, `Train`, and `ReservationSystem` to organize data and operations.When a confirmed ticket is cancelled, the first passenger in the waiting queue should be offered the available seat.

Updated Project Approach and Architecture (2 pts)

(Describe your current approach, including system design, communication protocols, libraries used, etc. Max 300 words).

The proposed implementation is a menu-driven C++ console application using standard C++ libraries.The design is organized around three main classes: `Passenger` stores passenger details such as name, age, PNR, train number, seat number, and booking status; `Train` represents train and seat information; and `ReservationSystem` coordinates bookings, cancellations, searches, and displays.A linked list will store confirmed passenger records because records can be inserted and removed dynamically.A queue will maintain waiting passengers in FIFO order, and a stack will record cancelled-ticket details so the latest cancellation can be viewed first. During booking, the system will check seat availability and either confirm the reservation or enqueue the passenger.During cancellation, the record will be added to cancellation history and the first waiting passenger will be promoted if a seat is available. The proposal specifies a console interface; no network protocol or live railway API is required.

Tasks Completed (7 pts)

(Describe the main tasks that have been assigned and already completed. Max 250 words).

Task / Work Established	Team Member
Defined booking, cancellation, search, and waiting-list requirements	Anmol Pandey and Prakul Maithani
Selected linked list, FIFO queue, and LIFO stack for reservation operations	Anmol Pandey and Prakul Maithani
Planned OOP classes: Passenger, Train, and ReservationSystem	Anmol Pandey and Prakul Maithani
Outlined menu-driven console workflow and expected deliverables	Anmol Pandey and Prakul Maithani

Challenges / Roadblocks (7 pts)

(Describe the challenges faced or being faced and how you plan to solve them. Max 300 words).

A key challenge is coordinating the three data structures so that passenger records remain consistent.Like,cancelling a confirmed booking should free a seat, record the cancellation, and move the first waiting passenger into the confirmed list when appropriate.This will be addressed by centralizing these operations in the `ReservationSystem` class and testing each operation with small examples. Another challenge is preventing duplicate or invalid values and ensuring searches return the correct passenger.Input validation and a unique identifier for each passenger will help. Queue and stack ordering must also be checked carefully: the waiting list must follow FIFO,while cancellation history must follow LIFO.Finally, the console menu should handle invalid choies and cases such as an empty waiting list, no available seats, or cancellation .These cases will be covered through structured test scenarios.

Tasks Pending (7 pts)

(Describe the main tasks that still need to be completed. Max 250 words).

Task Pending	Team Member
Implement and validate Passenger, Train, and ReservationSystem classes	Anmol Pandey and Prakul Maithani
Implement linked-list storage for confirmed passengers	Anmol Pandey and Prakul Maithani
Implement FIFO waiting queue and LIFO cancellation stack	Anmol Pandey and Prakul Maithani
Connect booking/cancellation flow, search, and display functions	Anmol Pandey and Prakul Maithani
Run test cases and prepare documentation and demonstration	Anmol Pandey and Prakul Maithani

Project Outcome / Deliverables (2 pts)

(Describe the key outcomes / deliverables. Max 200 words).

The intended outcome is a working C++ Railway Reservation and Waiting List Management System.It will allow users to view train and seat information,book tickets, add passengers to a waiting list when seats are unavailable, cancel confirmed tickets and automatically promote the first waiting passenger when a seat becomes free.Users will also be able to search passenger details, display confirmed passenger records, view the waiting list,and inspect cancellation history. Deliverables are the C++ source code,project documentation, test cases, and a working demonstration. The project will demonstrate linked lists,queues, stacks, and basic OOP concepts practically.

Progress Overview (2 pts)

(Summarize how much of the project is done, what is behind schedule, and what is ahead of schedule. Max 200 words).

The project has established the problem statement, core features, data-structure choices, planned classes, and development milestones.the code-level progress is yet to be completed with enhancements on our side. The main development work is to implement the classes and data structures,connect them to the booking and cancellation workflow,and test cases.The project should prioritize correct seat allocation and FIFO waitinglist promotion before the menu and preparing the final demo. Progress wil be tracked by marking each feature as completed only after it has been implemented and tested.

Codebase Information (2 pts)

(Repository link, branch, and information about important commits).

Repository link: https://github.com/incalculable04/railway-management-pbl-project
Branch: main
Important commits:

Initial project structure – Created the basic C++ project files and menu-driven program structure.
Add OOP class definitions – Added the Passenger, Train, and ReservationSystem classes.
Implement confirmed passenger linked list – Added linked-list operations to manage confirmed reservations.
Implement waiting queue and cancellation stack – Added FIFO waiting-list management and LIFO cancellation history.
Integrate booking and cancellation – Connected ticket booking, cancellation, seat allocation, and waiting-list promotion.
Add passenger search and sorting – Implemented PNR-based passenger searching and train sorting by fare or seat availability.
Add input validation and test cases – Tested booking, cancellation, invalid PNRs, full trains, and waiting-list promotion.

Testing and Validation Status (2 pts)

(Provide information about any tests conducted).

Test Type	**Status**	**Notes**
Ticket booking	Completed	Confirm booking when a seat is available.
Full train / waiting list	Completed	Add passenger to queue when no seats remain.
Cancellation	Completed	Remove confirmed passenger and record cancellation.
FIFO promotion	Planned	First waiting passenger receives a freed seat.
Cancellation history	Planned	Most recent cancellation appears first.
Invalid PNR / menu input	Planned	Show a suitable message without corrupting records.

Deliverables Progress (2 pts)

(Indicate whether each deliverable is completed, in progress, or pending).

Deliverable	**Status based on available source**
Project scope and feature plan	Completed
OOP class implementation	confirmed
Confirmed passenger linked list	Planned
Waiting-list queue	Planned
Cancellation-history stack	Planned
Booking/cancellation integration	Planned
Testing and final demonstration	Planned